

Radiology: infection control

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I. INTRODUCTION: RISKS IN DENTAL RADIOLOGY

Dental healthcare personnel and patients are routinely exposed to a variety of infectious agents. The nature of dental radiographic procedures—which involve close contact with oral fluids, mucous membranes, and respiratory droplets—significantly increases the risk of transmission for:

- Tuberculosis, HIV, Herpes viruses, Upper respiratory infections, Hepatitis viruses (A to E)

The primary goal of infection control in radiology is to prevent cross-contamination in all directions:

- From **patient to staff**
- From **staff to patient**
- From **patient to patient**

II. SOURCES OF CROSS-CONTAMINATION IN RADIOGRAPHY

Cross-contamination can occur through multiple pathways during radiographic procedures:

1. Operator Hand Contamination

Hands may become contaminated through contact with:

- The patient's mouth
- Saliva-contaminated films or film-holding devices

Following this, the operator may adjust the X-ray tube head, control panel, or other equipment, spreading contamination.

2. Darkroom Contamination

Handling digital sensors or opening contaminated film packets in the darkroom can transfer pathogens to processing equipment and other surfaces.

III. RESPONSIBILITY FOR INFECTION CONTROL

The dentist holds the ultimate responsibility to:

- Establish and enforce infection control protocols
- Minimize or eliminate cross-contamination risks
- Educate all team members on proper procedures
- Ensure compliance with standard precautions

IV. KEY STEPS IN RADIOGRAPHIC INFECTION CONTROL

1. Apply Standard Precautions

- Also known as universal precautions
- Treat all human blood and saliva as potentially infectious for HIV, hepatitis B, and other bloodborne pathogens
- Apply to all patients, regardless of known infection status

2. Wear Personal Protective Equipment (PPE)

- Hand Hygiene:

Wash hands with plain or antimicrobial soap before and after patient contact. Alcohol-based hand rubs are also effective.

- Gloves:

Wear disposable gloves during all radiographic procedures and when handling contaminated items. Change gloves between patients and if torn.

- Protective Clothing:

Wear a disposable gown or laboratory coat to protect skin and clothing from contamination.

3. Disinfect and Cover Clinical Contact Surfaces

Clinical contact surfaces are surfaces that may be touched by contaminated gloves, instruments, or directly by oral fluids. These include:

- X-ray machine and control panel
- Beam alignment device
- Dental chair and headrest
- Protective apron and thyroid collar
- Chair-side computer
- Surfaces where receptors are placed

Procedure:

- Clean and disinfect all surfaces between patients using intermediate- to low-level disinfectants
- Cover disinfected surfaces with plastic barriers (clear plastic wrap or bags)
- Replace barriers after each patient or if damaged

4. Sterilize Non-Disposable Instruments

- Film-holding instruments are classified as semi critical items (contact mucous membranes but do not penetrate soft tissue or bone)
- Sterilize using steam under pressure (autoclave) after each use

5. Use Barriers with Digital Sensors

- Digital sensors cannot be heat-sterilized

- Use single-use plastic barriers to protect sensors from saliva and blood during intraoral placement

- Change barrier for each patient

6. Prevent Contamination of Processing Equipment

Darkroom Protocol:

1. Remove contaminated gloves before entering darkroom
2. Place two clean towels on the work surface
3. Place container of contaminated film packets on first towel
4. Open packets and drop clean films onto second towel
5. Discard packaging on first towel
6. Only clean films should enter processing solutions

V. VISUAL GUIDE TO INFECTION CONTROL PRACTICES





15 10 Film holding instrument with barrier wrapping to protect



40-40 Method for removing film from packet without towel



VI. SPECIAL CONSIDERATIONS FOR PANORAMIC & CEPHALOMETRIC UNITS

- Panoramic Units:

Clean chin rest and handgrips with low-level disinfectant. Consider disposable biteblocks. Wipe head-positioning guides, control panel, and exposure switch with disinfectant-moistened towel.

- Cephalometric Units:

Clean and disinfect ear posts, brackets, forehead support, and nasion pointer. May also be covered with plastic barriers.

VII. POST-PROCEDURE CLEANUP

After completing exposures:

1. Remove and discard all plastic barriers
2. Disinfect all clinical contact surfaces
3. Disinfect protective apron
4. Replace barriers for next patient
5. Process films using aseptic darkroom technique

VIII. SUMMARY: THE INFECTION CONTROL CHAIN

Effective infection control in dental radiology requires breaking the chain of contamination at every step:

1. Protect the operator with PPE
2. Protect the patient with clean equipment and barriers
3. Protect the environment with disinfection and containment
4. Protect the image with aseptic processing techniques

By following standardized protocols, dental professionals can ensure safe, high-quality radiographic care while minimizing infection risks for all.